Azimuth Multichannel SAR Imaging Algorithm Based on Spectral Extrapolation and Image Fusion

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***Abstract*— The azimuth multichannel synthetic aperture radar (SAR) technique can simultaneously achieve high resolution and wide swath, which is important for disaster management, sea and land traffic observation, and environmental monitoring. However, nonuniform sampling leads to multichannel imaging quality degradation such as serious ghost targets or deterioration of signal-to-noise ratio (SNR). In this article, a multichannel SAR imaging algorithm is proposed. The key is to perform spectral selection and extrapolation based on the time–frequency characteristics of multichannel signals under nonuniform sampling conditions to obtain subimages with different ghost distributions. By implementing subimage fusion, the final imaging result can be obtained. Simulations and satellite real data experiments are conducted to verify the effectiveness of the proposed algorithm. The imaging performances of the proposed algorithm in terms of ghost-to-real target ratio (GRTR), and SNR are investigated with respect to the pulse repetition frequency (PRF). Moreover, simulation results also demonstrate that the proposed algorithm exhibits a more robust and consistent overall performance than existing methods.**

***Index Terms*— Azimuth multichannel synthetic aperture radar (SAR), image fusion, periodic nonuniform sampling, spectral extrapolation.**

I. Introduction

THE employment of synthetic aperture radar (SAR) in both civil and military domains has garnered global interest, particularly focusing on high-resolution and wide-swath (HRWS) imaging [1]. Azimuth multichannel SAR with multiple receiving antennas along the flight direction can overcome the minimum antenna area limitation [2] by increasing spatial sampling to compensate for temporal undersampling, enabling wide-swath observation while maintaining high-resolution imaging. For each receiving channel of an azimuth multichannel SAR system, the received echoes are undersampled, which leads to Doppler spectrum aliasing. The key to azimuth multichannel SAR imaging lies in the joint processing of multichannel data to suppress ghost targets caused by Doppler spectral aliasing. In this article, this goal is achieved through spectral extrapolation and image fusion.

There are several classical methods for multichannel SAR imaging. The digital beamforming (DBF) algorithm [3] uses the inverse matrix of the transfer matrix of a multichannel SAR system to filter out the aliasing components in the multichannel signals and reconstruct the aliasing-free spectrum. However, due to system design limitations and actual flight conditions, the pulse repetition frequency (PRF) often fails to satisfy the uniform sampling condition. Since the DBF filter significantly amplifies the noise power, its performance is seriously affected when highly nonuniform sampling occurs. The space time adaptive processing (STAP) algorithm [4], [5] applies the spatial beamforming technique to the multichannel signal reconstruction by designing a suitable spatial filter to align the zero-trapping position with the spectral aliasing components to filter out the spectral aliasing components, and then finally rearrange the retained spectral components to obtain the complete spectrum. The max-signal algorithm [6], the max signal-to-ambiguity-plus-noise ratio (SANR) algorithm [6], and the minimum mean square error (MMSE) algorithm [7] use different optimization criteria to construct the loss function and then use the optimal solution of the corresponding loss function to construct the optimal weighting matrix, so as to complete the optimal estimation of the signal spectrum. These algorithms can effectively improve the signal-to-noise ratio (SNR), but such algorithms are limited in their ability to suppress spectral aliasing components. The improved DBF (IDBF) algorithm [8], Capon-based algorithm [9], and improved MMSE algorithm [10] improve the SNR by changing the equivalent Doppler bandwidth of the reconstructed signal and discarding the Doppler components of the signal that have less energy, thus reducing the noise introduced in the reconstructed bandwidth. The minimum-energy-criterion (MEC) algorithm [11] and the improved MEC algorithm [12] consider the correlation between the azimuth frequencies of the multichannel signals, construct a new multichannel aliased signal model, and reconstruct the signal spectrum without degrading the SNR by solving the MEC optimization problem.

This article proposes an algorithm for multichannel SAR imaging under nonuniform sampling conditions. The key of the algorithm lies in establishing an azimuth multichannel SAR signal model that conforms to the periodic nonuniform

Manuscript received 9 December 2023; revised 7 April 2024 and 16 June 2024; accepted 31 July 2024. Date of publication 12 August 2024; date of current version 21 August 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62271031 and in part by the National Key Laboratory Foundation of Microwave Imaging Technology. *(Corresponding author: Yukun Guo.)*

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Digital Object Identifier 10.1109/TGRS.2024.3441693

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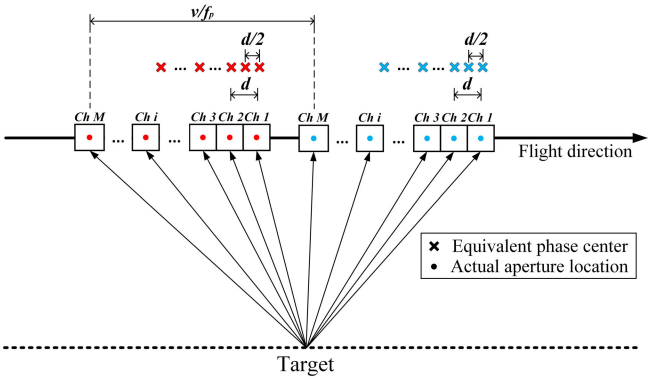


Fig. 1. Diagram of multichannel SAR system sampling.

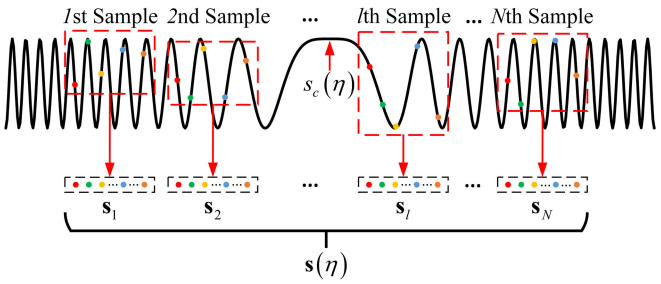


Fig. 2. Periodic nonuniform sampling of a chirp signal. Dots in the red dashed box indicate M sampled values from the same sample, and dots of the same color indicate sampled values of the same channel at different sampling moments.

sampling characteristics. Then, based on this model, the causes of ghost targets are explained from the perspective of time–frequency analysis, and finally, spectral extrapolation and image fusion are used to effectively suppress the ghost targets caused by nonuniform sampling.

The article is structured as follows. In Section II, based on periodic nonuniform sampling, the aliasing characteristics of the Doppler spectrum of the azimuth multichannel SAR signals are analyzed, and the relationship between the ghost targets and the spectral structure is given. In Section III, a multichannel SAR imaging algorithm based on spectral extrapolation and image fusion is proposed, whereby ghost targets are made to appear at different locations in the different subimages by extrapolating different subspectrums, and all ghost targets are further suppressed by image fusion. In Section IV, the performance of the proposed algorithm is compared with various algorithms (including DBF algorithm, IDBF algorithm, and STAP algorithm, etc.) using simulated and real multichannel data. In Section V, a conclusion is drawn.

II. Azimuth Multichannel SAR Signal Characterization

Fig. 1 illustrates an M-channel SAR system sampled twice consecutively along the flight direction and "Ch i" denotes the ith receiving channel. All channels are distributed along the flight direction and the spacing between neighboring channels is d. It is assumed that the first channel transmits the signal with low PRF which is denoted as f_p, and all channels receive the signal reflected from the target. The signal received by any channel can be regarded as emitted and received from its equivalent phase center (EPC) [3]. The EPC is approximately located at the midpoint of the actual transmitting and receiving channels. The spacing between adjacent EPCs is $d/2$.

In this section, the azimuth spectral characteristics of multichannel SAR signals are first investigated based on the periodic nonuniform sampling, and then the causes of ghost targets in multichannel SAR imaging are explained.

A. Signal Model Based on Periodic Nonuniform Sampling

The signals of each receiving channel can be viewed as discrete samples of the continuous signal $s_c(\eta)$ reflected from the target. As Fig. 1 shows, the sampling mode of azimuth multichannel SAR has a periodic nonuniform characteristic when $v/f_p \neq d/2$. Assuming that the SAR system receives N pulse echoes, the sampling moment of lth pulse for the mth channel is

$$\begin{aligned}\eta_{m,l} &= (l-1)\cdot\frac{1}{f_p} + (m-1)\cdot\frac{d}{2v} \\ &= \frac{[(l-1)\cdot M + (m-1) - r_m]}{M\cdot f_p}\end{aligned} \tag{1}$$

in which

$$\begin{aligned}r_m &= (m-1)\cdot\left[1 - M\cdot f_p\cdot d/(2v)\right] \\ l &= 1,2,\ldots,N, \quad m = 1,2,\ldots,M\end{aligned} \tag{2}$$

where r_m characterizes the extent to which periodic nonuniform sampling deviates from uniform sampling.

The signals of each receiving channel are rearranged as shown in Fig. 2 to obtain a periodic nonuniform sampling sequence $\mathbf{s}(\eta)$. $\mathbf{s}_l$ is defined as

$$\mathbf{s}_l = \left[s_c\left(\eta_{1,l}\right), s_c\left(\eta_{2,l}\right), \ldots, s_c\left(\eta_{M,l}\right)\right], \quad l = 1,2,\ldots,N. \tag{3}$$

The relationship between $\mathbf{s}_l$ and $\mathbf{s}(\eta)$ is

$$\mathbf{s}(\eta) = [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_l, \ldots, \mathbf{s}_N]. \tag{4}$$

Based on the properties of periodic nonuniform sampling sequences [13], [14], [15], the Doppler spectrum of $\mathbf{s}(\eta)$ can be expressed as (see Appendix A for detailed derivation)

$$S\left(f_\eta\right) = f_p\cdot\sum_{k=-\infty}^{+\infty}\sum_{m=1}^{M} F_{k,m}\left(f_\eta\right)\cdot S_c\left(f_\eta - k\cdot f_p\right) \tag{5}$$

where f_η denotes Doppler frequency. $S(f_\eta)$ and $S_c(f_\eta)$ denote the spectrum of $\mathbf{s}(\eta)$ and $s_c(\eta)$, respectively. $F_{k,m}(f_\eta)$ denotes the weighting factor of the frequency-shift component $S_c(f_\eta - k\cdot f_p)$.

Under uniform sampling condition, $f_p = 2\cdot v/(M\cdot d)$, and (5) can be simplified as

$$S\left(f_\eta\right) = f_p\cdot\sum_{k=-\infty}^{+\infty}\sum_{m=1}^{M}\exp\left\{-j\cdot\left[\frac{k\cdot(m-1)\cdot 2\pi}{M}\right]\right\} \cdot S_c\left(f_\eta - k\cdot f_p\right). \tag{6}$$

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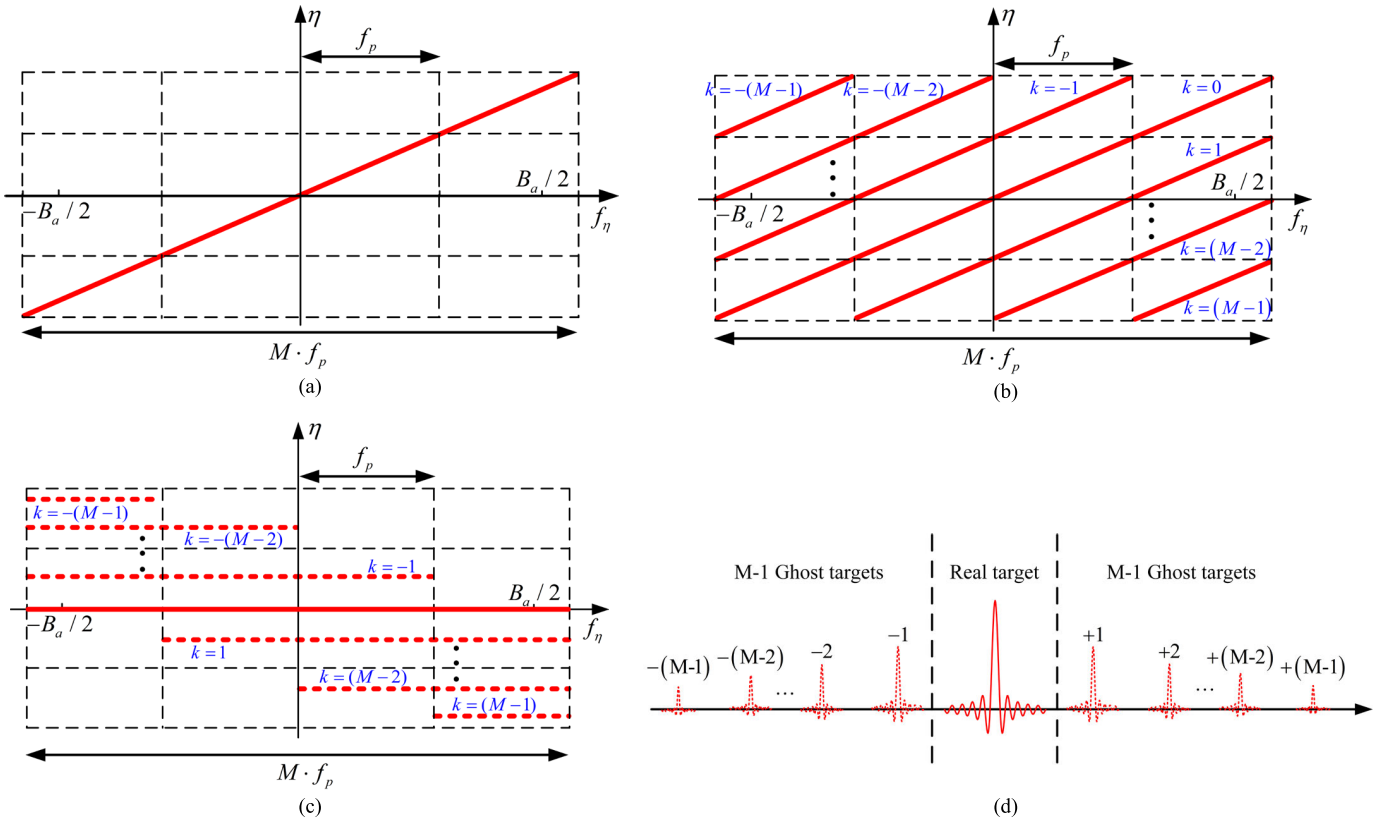


Fig. 3. Time–frequency characteristics of signals and distribution of ghost targets under bandlimited condition. (a)–(c) Time–frequency characteristics corresponding to $S_c(f_\eta)$, $S(f_\eta)$, and $\hat{S}(f_\eta)$, respectively. (d) Distribution of real and ghost targets along azimuth direction.

Considering

$$\sum_{m=1}^{M} \exp\left\{-j \cdot \left[\frac{k \cdot (m-1) \cdot 2\pi}{M}\right]\right\} = \begin{cases} M, & k = l \cdot M \\ 0, & k \neq l \cdot M. \end{cases} \tag{7}$$

Equation (6) transfers to

$$S(f_\eta) = M \cdot f_p \cdot \sum_{l=-\infty}^{+\infty} S_c(f_\eta - l \cdot M \cdot f_p). \tag{8}$$

Equation (8) is exactly the spectrum of the signal obtained by uniform sampling at the sampling rate $M \cdot f_p$ [16] and represents the spectrum of the azimuth multichannel SAR under uniform sampling conditions, which can be regarded as a special case of periodic nonuniform sampling conditions.

B. Causes of Ghost Targets Under Bandlimited Condition

Conventional multichannel signal reconstruction algorithms only consider the Doppler spectrum in the frequency range $[-M \cdot f_p/2, M \cdot f_p/2]$. In this section, the cause of ghost targets under bandlimited condition is analyzed.

Fig. 3(a) shows the time–frequency characteristics of $s_c(\eta)$. B_a denotes the 3 dB Doppler bandwidth which is smaller than $M \cdot f_p$. According to (4), $S(f_\eta)$ consists of $S_c(f_\eta)$ and multiple frequency-shift components $S_c(f_\eta - k \cdot f_p)$ in which k is a nonzero integer. These components are distributed in the frequency range $[-M \cdot f_p/2, M \cdot f_p/2]$ and can be obtained by shifting $S_c(f_\eta)$ along the Doppler frequency axis by an integer multiple of f_p. There are $(M-1)$ frequency-shifted components on each side of $S_c(f_\eta)$ as shown in Fig. 3(b).

After processing $\mathbf{s}(\eta)$ using a matched filter suitable for $s_c(\eta)$ (see Appendix B for detailed derivation), the phase of $S(f_\eta)$ is compensated to obtain $\hat{S}(f_\eta)$, as shown in Fig. 3(c), where the solid and dashed lines represent the results after phase compensation of $S_c(f_\eta)$ and $S_c(f_\eta - k \cdot f_p)$, respectively. The phase compensated results of $S_c(f_\eta)$ and $S_c(f_\eta - k \cdot f_p)$ are denoted as $\hat{S}_c(f_\eta)$ and $\hat{S}_c(f_\eta - k \cdot f_p)$, respectively. The time–frequency curves of each spectral component are parallel to the Doppler frequency axis.

Transforming $\hat{S}(f_\eta)$ to the time domain by azimuth inverse Fourier Transform, the focusing result $\hat{s}(\eta)$ can be obtained, as shown in Fig. 3(d) where the solid line represents the real target generated by $\hat{S}_c(f_\eta)$, and the dashed line represents the $2 \cdot (M-1)$ ghost targets generated by $\hat{S}_c(f_\eta - k \cdot f_p)$ on both sides of the real target.

The distance of the ghost targets offset in azimuth direction with respect to the real target is (see Appendix B for detailed derivation)

$$P_{l,k} = (l - k/M) \cdot \alpha_s \cdot v \cdot T_a \tag{9}$$

where

$$\alpha_s = M \cdot f_p / B_a$$
$$l = 0, 1, \quad k = 1, 2, \ldots, M-1. \tag{10}$$

When $l = 0$, (9) denotes the offset features of the ghost targets on the left side of the real target, and when $l = 1$, (9)

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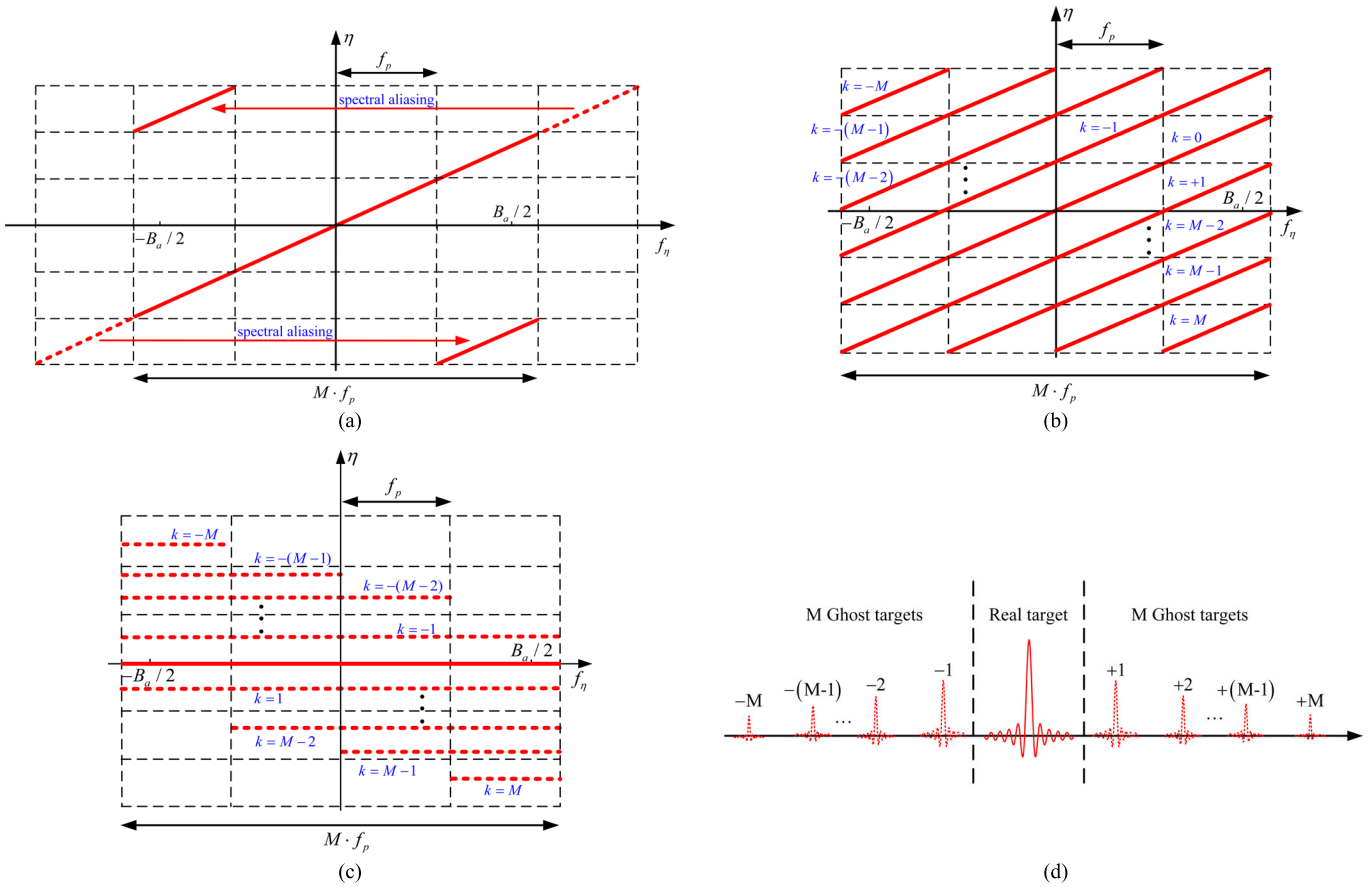


Fig. 4. Time–frequency characteristics of signals and distribution of ghost targets under nonbandlimited condition. (a)–(c) Time–frequency characteristics corresponding to $S_c(f_\eta)$, $S(f_\eta)$, and $\hat{S}(f_\eta)$, respectively. (d) Distribution of real and ghost targets along azimuth direction.

denotes the offset features of the ghost targets on the right side of the real target.

C. Additional Ghost Targets Under Nonbandlimited Condition

The actual SAR signal is nonbandlimited and modulated by the antenna pattern. Conventional multichannel imaging algorithm is developed for bandlimited signal. However, the spectrum outside $[-M \cdot f_p/2, M \cdot f_p/2]$ leads to an increasing number of aliased spectral components, which generate more ghost targets [17].

Fig. 4 considers the spectrum of the signal within $[-(M+2) \cdot f_p/2, (M+2) \cdot f_p/2]$ and illustrates how the spectrum outside $[-M \cdot f_p/2, M \cdot f_p/2]$ generates the additional ghost targets. Fig. 4(a) shows the time–frequency characteristics of $S_c(f_\eta)$. It can be seen that spectral aliasing causes the spectrum outside $[-M \cdot f_p/2, M \cdot f_p/2]$ which is denoted by red dashed line to fold into the $[-M \cdot f_p/2, M \cdot f_p/2]$. Thus, $S(f_\eta)$ contains more spectral components in $[-M \cdot f_p/2, M \cdot f_p/2]$ compared with the bandlimited condition as shown in Fig. 3(b). Fig. 4(b) and (c) show the time–frequency characteristics of $S(f_\eta)$ and $\hat{S}(f_\eta)$, respectively. The final distribution of ghost targets is shown in Fig. 4(d), where M ghost targets appear on each side of the real target. In the case of an infinite actual signal bandwidth, the number of ghost targets is also infinite theoretically.

III. Imaging Algorithm Based on Spectral Extrapolation and Image Fusion

Based on the analysis in Section II, a multichannel SAR imaging algorithm using spectral extrapolation and image fusion is proposed in this section, which first suppresses partial ghost targets by spectral selection and extrapolation, then constructs a ghost target map based on the amplitude difference of the ghost targets in different subimages, and finally fuse these subimages to get the final image in which all the ghost targets are suppressed.

In the following, the spectral selection and extrapolation under bandlimited and nonbandlimited conditions are first elaborated, then the processing of image fusion is described, and finally, the workflow of the proposed algorithm is given.

A. Spectral Selection and Extrapolation

1) Bandlimited Condition: Dividing the spectrum $\hat{S}(f_\eta)$ equally into M subspectrums within $[-M \cdot f_p/2, M \cdot f_p/2]$, these subspectrums are denoted in the order from left to right as $\hat{S}_n(f_\eta), n = 1, 2, \ldots, M$. $\hat{S}_n(f_\eta)$ corresponds to azimuth frequencies $[-M \cdot f_p/2 + (n-1) \cdot f_p, -M \cdot f_p/2 + n \cdot f_p]$ and contains spectral components $\hat{S}_c(f_\eta - k_n \cdot f_p)$, in which the maximum value $k_n^{\max}$ and the minimum value $k_n^{\min}$ of the index k_n satisfy

$$\begin{cases} M \cdot f_p/2 + k_n^{\min} \cdot f_p = -M \cdot f_p/2 + n \cdot f_p \\ -M \cdot f_p/2 + k_n^{\max} \cdot f_p = -M \cdot f_p/2 + (n-1) \cdot f_p. \end{cases} \tag{11}$$

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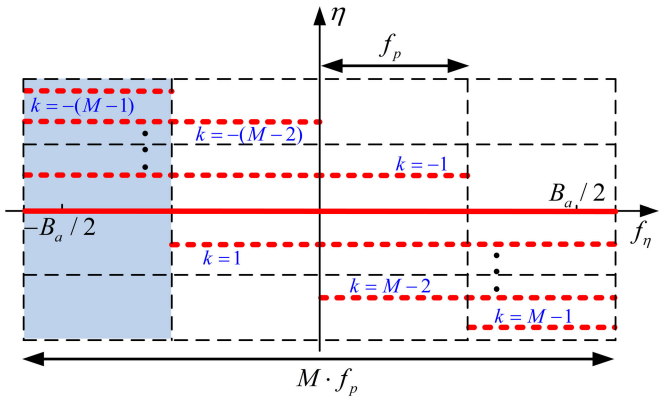


Fig. 5. Distribution of spectral components in subspectrum $\hat{S}_1(f_\eta)$ under bandlimited condition.

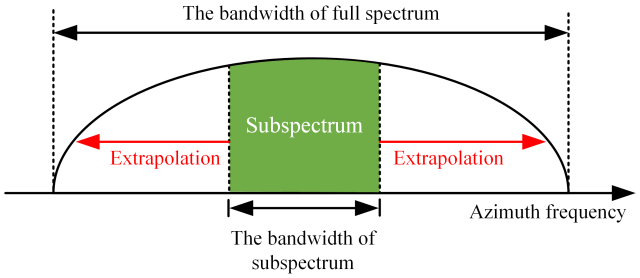


Fig. 6. Diagram of spectral extrapolation. The green area indicates the selected subspectrum.

From (11), it can be known that $k_n^{\min}=n-M$ and $k_n^{\max}=n-1$. Thus, the subspectrum $\hat{S}_n(f_\eta)$ contains the spectral components $\hat{S}_c(f_\eta-k_n\cdot f_p)$ in which $k_n=n-M,n-M+1,\ldots,n-1$. Fig. 5 shows the distribution of spectral components using $\hat{S}_1(f_\eta)$ as an example. As can be seen from the blue shading, $\hat{S}_1(f_\eta)$ contains the spectral components labeled as $k=-(M-1),-(M-2),\ldots,0$ which correspond to the ghost targets labeled as $-(M-1),-(M-2),\ldots,-1$ and the real target in Fig. 3(d).

After transforming $\hat{S}_n(f_\eta)$ to the time domain by azimuth inverse Fourier Transform, a new image can be obtained. Compared to the original image obtained based on the full spectrum, the newly obtained image contains one real target and $M-1$ ghost targets, while the other $M-1$ ghost targets are suppressed, but it suffers from azimuth resolution deterioration due to the reduction of the azimuth bandwidth [18].

Spectral extrapolation with weighted time-domain energy measure [19] is an effective method to preserve the resolution and has been used in SAR image processing [20], [21]. The process of spectral extrapolation is illustrated in Fig. 6. In this section, the bandwidth of subspectrum $\hat{S}_n(f_\eta)$ is f_p, and the bandwidth of full spectrum $\hat{S}(f_\eta)$ is $M\cdot f_p$.

Based on the above analysis, by selecting and extrapolating the subspectrum, target information within the selected subspectrum can be retained without reducing the azimuth resolution. The subimage $\hat{s}_n(\eta)$ obtained by extrapolating the subspectrum $\hat{S}_n(f_\eta)$ contains the real target and the ghost targets labeled as from $n-M$ to $n-1$, while the other ghost targets are suppressed. So as n varies, the locations of the ghost targets in different subimages are different. Since certain ghost targets are suppressed in certain subimages, it is possible to suppress all the ghost targets by fusing these subimages.

Denoting the 1-D signal of $\hat{s}(\eta)$, $\hat{s}_n(\eta)$, and $\hat{S}_n(f_\eta)$ at a certain range gate R as $\hat{s}(\eta;R)$, $\hat{s}_n(\eta;R)$, and $\hat{S}_n(f_\eta;R)$, the objective function of spectral extrapolation is [20], [21]

$$\begin{aligned}&\min\left\langle\hat{s}_n(\eta;R),\mathbf{W}^{-1}\hat{s}_n(\eta;R)\right\rangle\\&\text{s.t. }\hat{S}_n\left(f_\eta;R\right)=\mathbf{F}_n\hat{s}_n(\eta;R)\end{aligned}\tag{12}$$

where $\mathbf{W}=\mathrm{diag}[\hat{s}(\eta;R)\hat{s}^*(\eta;R)]$. The symbol diag[·] means retaining the main diagonal elements of the matrix and zeroes the other elements of the matrix, $*$ represents conjugate transpose operator, and $\langle\cdot,\cdot\rangle$ denotes the standard inner product in Hilbert space. $\mathbf{F}_n$ is a Fourier Transform matrix which reflects the relationship between the original image $\hat{s}(\eta;R)$ and the nth selected subspectrum $\hat{S}_n(f_\eta;R)$, and its size is $N\times N_{\text{azi}}$, where N_{azi} is the number of azimuth samples of $\hat{s}(\eta;R)$. Based on the analysis in Section II, it is known that $N_{\text{azi}}=M\cdot N$, and the element in the i_1th row and i_2th column of matrix $\mathbf{F}_n$ is

$$\begin{aligned}&\mathbf{F}_n(i_1,i_2)\\&=\exp\left\{\frac{-j\cdot2\pi\cdot\left[i_1+(n-1)\cdot N-N_{\text{azi}}/2-1\right]\cdot(i_2-1)}{N_{\text{azi}}}\right\}\\&i_1\in\{1,2,\ldots,N\},\quad i_2\in\{1,2,\ldots,N_{\text{azi}}\},\quad j=\sqrt{-1}.\end{aligned}\tag{13}$$

Equation (12) imposes constraints on $\hat{s}_n(\eta;R)$ from both the time domain and the frequency domain, making the estimated result $\hat{s}_n(\eta;R)$ similar to $\hat{s}(\eta;R)$ in the time-domain waveform, matching the spectral range of $\hat{s}(\eta;R)$, and ensuring that the ghost target information not included in $\hat{S}_n(f_\eta;R)$ does not appear in $\hat{s}_n(\eta;R)$. According to the projection theorem [22], the relationship between $\hat{s}(\eta;R)$ and $\hat{s}_n(\eta;R)$ is

$$\hat{s}_n(\eta;R)=\mathbf{W}\mathbf{F}_n^*\left(\mathbf{F}_n\mathbf{W}\mathbf{F}_n^*\right)^{-1}\mathbf{F}_n\hat{s}(\eta;R).\tag{14}$$

2) Nonbandlimited Condition: Under nonbandlimited condition, more than M spectral components $\hat{S}_c(f_\eta-k_n\cdot f_p)$ are contained in subspectrum $\hat{S}_n(f_\eta)$ when the subspectrum is still obtained by dividing the full spectrum equally. Considering the spectrum within $[-(M+2)\cdot f_p/2,(M+2)\cdot f_p/2]$, the maximum value $k_n^{\max}$ and the minimum value $k_n^{\min}$ of the index k_n satisfy

$$\begin{cases}(M+2)\cdot f_p/2+k_n^{\min}\cdot f_p=-M\cdot f_p/2+n\cdot f_p\\-(M+2)\cdot f_p/2+k_n^{\max}\cdot f_p=-M\cdot f_p/2+(n-1)\cdot f_p.\end{cases}\tag{15}$$

From (15), $k_n^{\min}=n-M-1$ and $k_n^{\max}=n$. Therefore, the subspectrum $\hat{S}_n(f_\eta)$ contains the spectral component $\hat{S}_c(f_\eta-k_n\cdot f_p)$ in which $k_n=n-M-1,n-M,\ldots,n$. Fig. 7 illustrates the increase in the number of spectral components contained in the selected subspectrum under nonbandlimited condition using $\hat{S}_1(f_\eta)$ as an example. It can be seen that compared with Fig. 5, the spectral component labeled as $k=-M$ and $k=1$ are contained in $\hat{S}_1(f_\eta)$. Compared with the bandlimited condition, there are less

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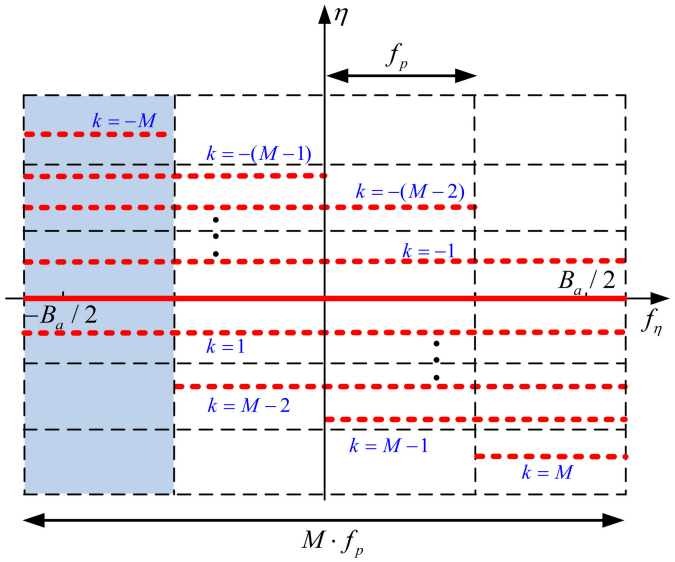


Fig. 7. Distribution of spectral components in subspectrum $\hat{S}_1(f_\eta)$ under nonbandlimited condition.

ghost targets suppressed in every subimage after spectral extrapolation because more spectral components are contained in all subspectrums. Besides, all of the subimages obtained after spectral extrapolation contain the ± 1 order ghost target because all the subspectrums contain $\hat{S}_c(f_\eta + f_p)$ and $\hat{S}_c(f_\eta - f_p)$.

Without the weighting of antenna azimuth pattern, $S_c(f_\eta)$ has a rectangular envelope, and matched filtering does not change the distribution of spectral energy. Therefore, the ratio of the energy of $\hat{S}_c(f_\eta - k_n \cdot f_p)$ to the energy of $\hat{S}_c(f_\eta)$ in $\hat{S}_n(f_\eta)$ without considering azimuth antenna pattern under bandlimited condition is

$$\gamma_1 = \frac{\int_{-M\cdot f_p/2+(n-1)\cdot f_p}^{-M\cdot f_p/2+n\cdot f_p} \left|S_c\left(f_\eta - k_n\cdot f_p\right)\right|^2 df_\eta}{\int_{-M\cdot f_p/2+(n-1)\cdot f_p}^{-M\cdot f_p/2+n\cdot f_p} \left|S_c\left(f_\eta\right)\right|^2 df_\eta} = \frac{\int_{-M\cdot f_p/2+(n-1)\cdot f_p}^{-M\cdot f_p/2+n\cdot f_p} A^2 df_\eta}{\int_{-M\cdot f_p/2+(n-1)\cdot f_p}^{-M\cdot f_p/2+n\cdot f_p} A^2 df_\eta} = 1 \tag{16}$$

where A denotes the amplitude of envelope of $S_c(f_\eta)$.

Considering the weighting of the antenna azimuth pattern, the ratio of the energy of $\hat{S}_c(f_\eta - k_n \cdot f_p)$ to the energy of $\hat{S}_c(f_\eta)$ in $\hat{S}_n(f_\eta)$ is

$$\gamma_2 = \frac{\int_{-M\cdot f_p/2+(n-1)\cdot f_p}^{-M\cdot f_p/2+n\cdot f_p} \left|W\left(f_\eta - k_n\cdot f_p\right)\right|^2 \cdot \left|S_c\left(f_\eta - k_n\cdot f_p\right)\right|^2 df_\eta}{\int_{-M\cdot f_p/2+(n-1)\cdot f_p}^{-M\cdot f_p/2+n\cdot f_p} \left|W\left(f_\eta\right)\right|^2 \cdot \left|S_c\left(f_\eta\right)\right|^2 df_\eta} = \frac{\int_{-M\cdot f_p/2+(n-1)\cdot f_p}^{-M\cdot f_p/2+n\cdot f_p} \left|W\left(f_\eta - k_n\cdot f_p\right)\right|^2 df_\eta}{\int_{-M\cdot f_p/2+(n-1)\cdot f_p}^{-M\cdot f_p/2+n\cdot f_p} \left|W\left(f_\eta\right)\right|^2 df_\eta} \tag{17}$$

where $W(\cdot)$ represents the antenna azimuth pattern weighting.

According to (17) and the Parseval's theorem [16], γ_2 changes as k_n varies which also means that the ratio of the amplitude of the same ghost target to that of the real target in different subimages does not keep the same. Thus, the detection of ghost targets can also be accomplished based on the difference of amplitude.

B. Image Fusion Based on Ghost Map

According to the above analysis, there are amplitude differences in the ghost target area of different subimages obtained by extrapolating different subspectra. The following describes how to use the amplitude difference to construct a ghost target map and finally obtain the final image in which all the ghost targets are suppressed.

For the convenience of expression, the nth subimage $\hat{s}_n(\eta)$ and the original image $\hat{s}(\eta)$ are rewritten in 2-D form as P_n and P, The values of P_n and P at position (a, r) are denoted as $P_n(a, r)$ and $P(a, r)$, respectively, in which a and r denote the azimuth and range positions of the pixel, respectively.

The ratio of the power of the two images P_n and P at position (a, r) after power normalization can be expressed as

$$R_n(a, r) = \frac{|P(a, r)|^2}{|P_n(a, r)|^2} \cdot \frac{Av\langle |P_n|^2 \rangle}{Av\langle |P|^2 \rangle} \tag{18}$$

where $|P(a, r)|$ and $|P_n(a, r)|$ denotes the amplitude of P and P_n at the position (a, r), respectively. $Av\langle\cdot\rangle$ means taking the average value of "$\cdot$," so $Av\langle|P_n|^2\rangle$ and $Av\langle|P|^2\rangle$ denotes the average power of P and P_n, respectively. The reason why the average power is used for normalization is to ensure that $R_n(a, r)$ is as close to 1 as possible at locations where the ghost target is weak, and as far above 1 as possible at locations that are greatly affected by the ghost target.

The binary map $B_n(a, r)$ can be obtained by performing threshold processing on (18)

$$B_n(a, r) = \begin{cases} 1, & R_n(a, r) > r_1 \text{ or } R_n(a, r) < r_2 \\ 0, & r_2 \le R_n(a, r) \le r_1 \end{cases} \tag{19}$$

where the threshold r_1 is generally taken as a real number greater than 1 and the threshold r_2 is generally taken as a real number between 0 and 1. According to (19), the area where the ghost target appears can be roughly detected.

However, due to the influence of speckle noise and thermal noise, some pixels may be misjudged. This problem can be solved by performing sliding window processing on the binary map [23]. If the number of nonzero pixels in the sliding window is lower than the threshold n_t set in advance, the values of all nonzero pixels in the sliding window are set to 0; otherwise, they are set to 1.

After obtaining M binary maps $B_n(a, r)$, the ghost target map $B(a, r)$ is constructed according to

$$B(a, r) = \begin{cases} 0, & \text{if all } B_n(a, r) = 0 \\ 1, & \text{else} \end{cases} \tag{20}$$

which indicates that once there exists any binary map $B_n(a, r)$ with a value of 1 at the position (a, r), the area is considered to be the place where the ghost target occurs.

According to the ghost target map $B(a, r)$, the value of the final image P_{final} at the position (a, r) can be generated as

$$P_{\text{final}}(a, r) = \begin{cases} P(a, r), & \text{if } B(a, r) = 0 \\ \delta \cdot P_{\text{select}}(a, r), & \text{if } B(a, r) = 1 \end{cases} \tag{21}$$

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where

$$P_{\text{select}}(a,r) = \min_n(|P_n(a,r)|) \tag{22}$$

$$\delta = \sqrt{|P_n|^2/|P|^2} \tag{23}$$

where $\min_n(|P_n(a,r)|)$ means the minimum amplitude value at the position (a,r) from M subimages where $B(a,r)=1$ is satisfied. The parameter δ denotes the square root of the power ratio between the subimage P_n corresponding to $P_{\text{select}}(a,r)$, and the original image P at position (a,r), which is used to ensure that the average value of the image power at the position before and after image fusion does not change much. Compared with the original image P, all ghost targets are significantly suppressed in the final image P_{final}, and the azimuth resolution is also maintained as much as possible.

The parameters r_1, r_2, n_t, and the size of sliding window mentioned above need to be compromised in terms of ghost target detection and image resolution preservation according to the actual situation.

C. Workflow of the Proposed Algorithm

Combining spectral selection and extrapolation and image fusion, a novel multichannel SAR imaging algorithm is proposed whose illustration is shown in Fig. 8.

Specific steps are as follows.

Step 1: The signals of each receiving channel reflected from the observation scene shown in Fig. 8(a) are rearranged to obtain a periodic nonuniform sampling sequence as described in Section II-A.

Step 2: Using traditional SAR imaging algorithms, such as the chirp-scaling algorithm [24], the periodic nonuniform sampling sequence constructed in Step 1 is processed to obtain a coarse imaging result containing ghost targets which are shown on Fig. 8(b). In Fig. 8(b), the cars on each side of the house are almost covered by the ghost targets from the house.

Step 3: Azimuth Fourier transform and spectral selection are performed on the coarse imaging result in Step 2 to obtain M subspectrums as shown in the red dashed box in Fig. 8.

Step 4: Spectral extrapolation and azimuth inverse Fourier transform are performed on each subspectrum to obtain the corresponding M subimages, in which the ghost targets at different locations are suppressed as shown in the blue dashed box in Fig. 8. As shown in Fig. 8(f1) and (f2), the influence of the ghost targets from the house on car weakens in the suppressed ghost target area.

Step 5: The subimages in Step 4 and the coarse imaging result in Step 2 are used to construct a ghost target map, and image fusion is completed based on the ghost target map to obtain the final image in which all the ghost targets are suppressed as shown in Fig. 8(g).

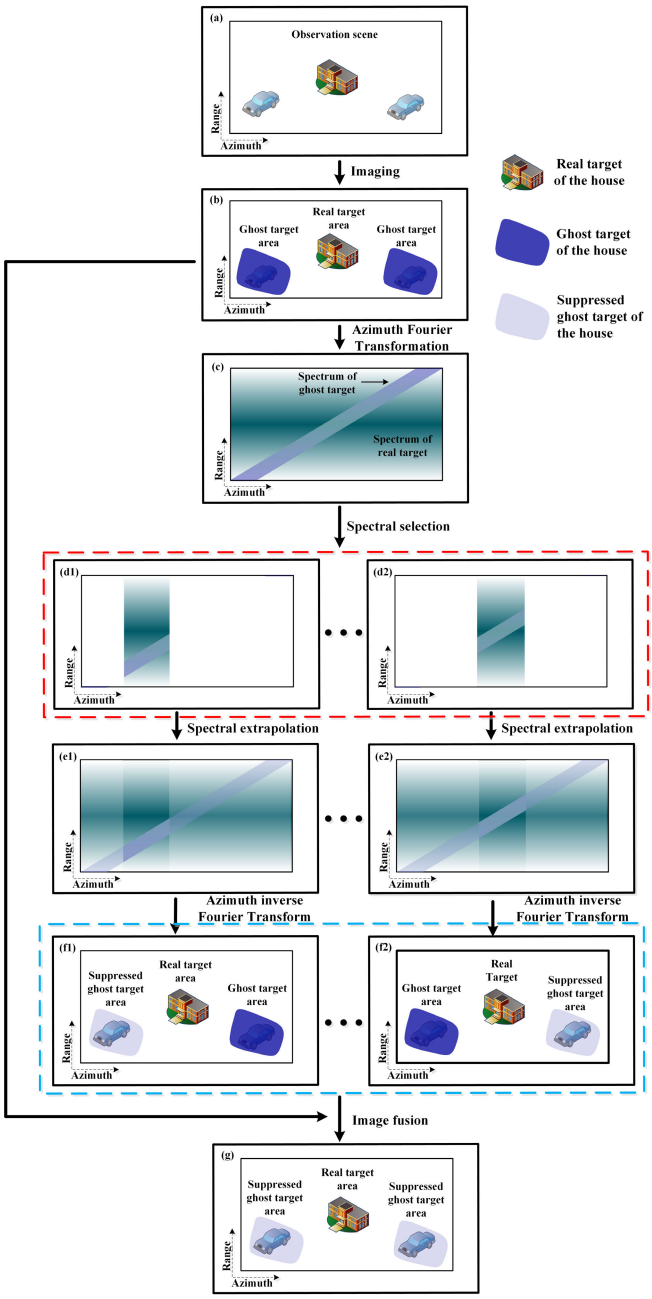


Fig. 8. Illustration of the proposed algorithm. (a) Denotes the observation scene. (b) Denotes the coarse imaging result. Only the ghost targets of the house are considered and the car is covered by the strong ghost target from the house in the ghost target area. (c) Denotes the azimuth spectrum of the coarse imaging result. (d_1) and (d_2) Red dashed box denote the subspectrums obtained by spectral selection. (e_1) and (e_2) Denote the results of spectral extrapolation based on the subspectrums in (d_1) and (d_2), respectively. (f_1) and (f_2) Blue dashed box denote the subimages obtained by spectral extrapolation and azimuth inverse Fourier transform, respectively. In the suppressed ghost target area, the influence of the ghost targets from houses in cars weakens, and the car becomes more visible. (g) Denotes the final imaging result of the proposed algorithm.

IV. Experiments and Analysis

In this section, simulation and satellite real-data experiments are conducted to validate the effectiveness of the proposed algorithm, as well as to compare and analyze the performance of the proposed algorithm with the DBF algorithm, the IDBF algorithm, the STAP algorithm, and the max-signal algorithm in terms of SNR and ghost-to-real target ratio (GRTR). GRTR is used to measure the prominence of the ghost target in the image and is defined as

$$\text{GRTR} = 20\log_{10}\frac{Av\langle\max(|I_G|)\rangle}{\max(|I_R|)} \tag{24}$$

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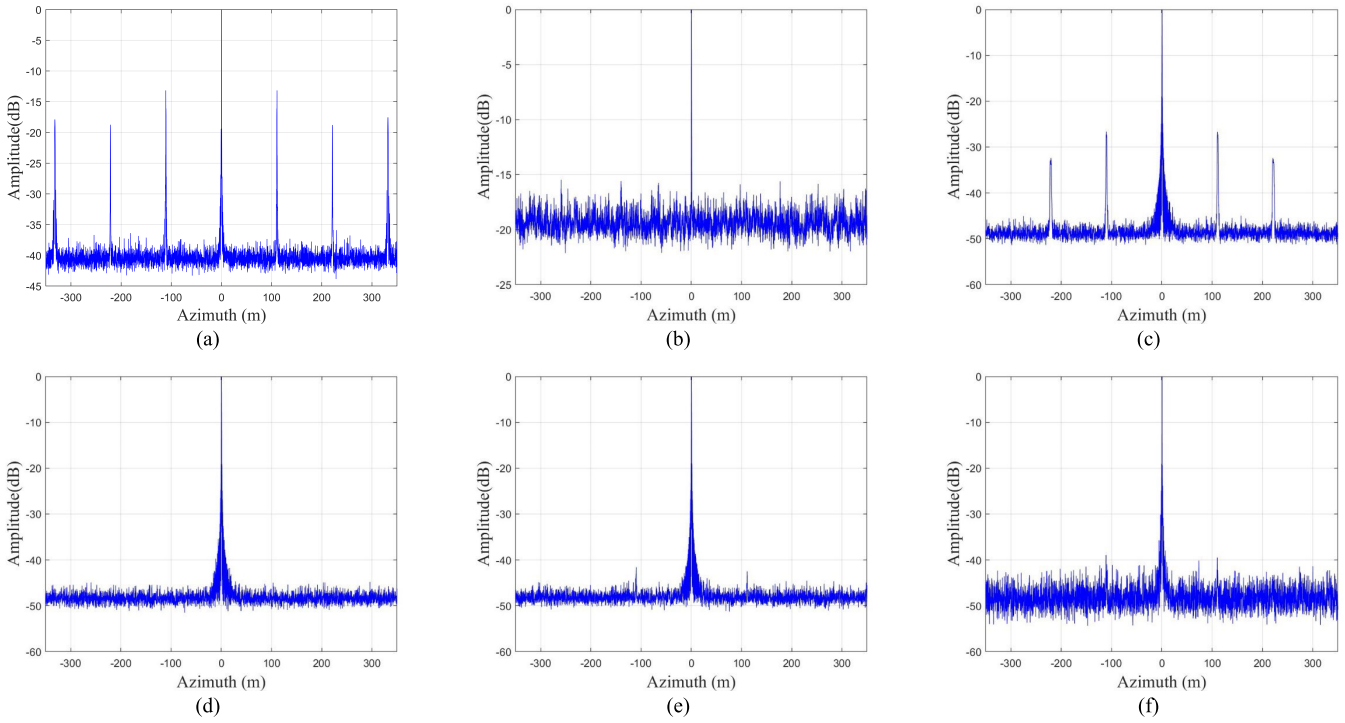


Fig. 9. Azimuth profile of different algorithms. (a) Periodic nonuniform sampling sequence processed by the chirp-scaling algorithm. (b) DBF algorithm. (c) Max-signal algorithm. (d) IDBF algorithm. (e) STAP algorithm. (f) Proposed algorithm.

where $\max(|I_R|)$ denotes the maximum magnitude of the real target in the image and $Av\langle\max(|I_G|)\rangle$ denotes the average of the peak magnitude of all the ghost targets.

A. Experiment for Simulated Multichannel Data

In this section, the proposed algorithm is first compared with several conventional multichannel imaging algorithms under bandlimited condition to verify the imaging performance in the frequency range of $[-M \cdot f_p/2, M \cdot f_p/2]$. Then, the proposed algorithm is verified for its ability to suppress additional ghost targets caused by the spectrum outside $[-M \cdot f_p/2, M \cdot f_p/2]$ under nonbandlimited condition.

1) Imaging Performance at a Single PRF Under Bandlimited Condition: The single-point target multichannel SAR signal is generated based on the parameters listed in Table I, and processed using the proposed algorithm and several conventional algorithms [3], [4], [6], [8]. The PRF is set to be 266 Hz, which satisfies the highly nonuniform sampling condition. Gaussian white noise is added to analyze the effect of different algorithms on the SNR and the SNR in the echo domain (hereafter denoted as SNR_{in}) is -25 dB. The considered azimuth bandwidth is set to be 1064 Hz which is consistent with the analysis in Section III-A.

Fig. 9 presents the azimuth profile of the results of different algorithms. Fig. 9(a) shows the imaging result of a periodic nonuniform sampling sequence processed by the chirp-scaling algorithm, where three ghost targets can be seen on each side of the real target. Besides, the distance between the first ghost target on the right and the real target is 110.7140 m, and the theoretical value is 110.5710 m, which indicates that (9) is basically correct. Fig. 9(b) shows the result of the DBF algorithm. It can be seen that ghost targets are effectively suppressed, but the noise level is significantly elevated compared with Fig. 9(a). This is because the DBF algorithm is designed with only the suppression of Doppler aliasing components, and the reconstruction filter indiscriminately amplifies the desired spectral components and the noise, especially in the case of highly nonuniform sampling (as in the case here of 266 Hz which satisfies the nearly coinciding PRF condition [6]), where the deterioration of the SNR becomes more pronounced. Fig. 9(c) shows the result of the max-signal algorithm. As can be seen, there is some reduction in noise levels compared with Fig. 9(a), but ghost targets are still evident. This is because the max-signal algorithm can maximize the energy of the signal and effectively reduce the noise energy, but it cannot effectively separate the Doppler aliasing component from the desired signal component, so that it has limited suppression

TABLE I
SYSTEM PARAMETERS FOR SIMULATED DATA

Parameter	Value
Carrier Frequency	9.6 GHz
Platform Height	5 km
Platform Velocity	200 m/s
Pulse Repetition Frequency	266 Hz/230 Hz
Channel Number	4
Channel Spacing	0.5 m
Antenna aperture length	0.5 m
Bandwidth	150 MHz
Sampling Rate	180 MHz
Pulse Duration	5 μs

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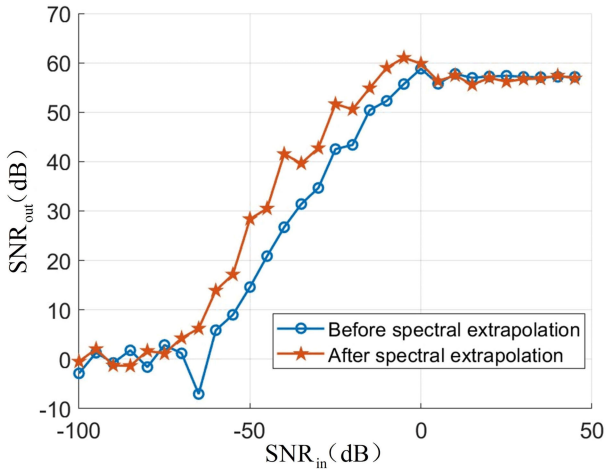


Fig. 10. SNR before and after spectral extrapolation.

TABLE II
EVALUATION OF IMAGING PERFORMANCE UNDER BANDLIMITED CONDITION

Algorithm	SNR (dB)	GRTR (dB)	Resolution (m)
DBF	15.3858	-52.8865	0.2109
IDBF	50.2465	-56.1609	0.2111
Max-signal	51.2516	-21.1438	0.2109
STAP	46.3006	-47.8706	0.2110
Proposed	55.3508	-53.9525	0.2267

ability for ghost targets. Fig. 9(d) shows the result of the IDBF algorithm which has a better SNR compared with Fig. 9(b). This is because the signal reconstruction bandwidth of the IDBF algorithm is smaller than that of the DBF algorithm, and less noise energy is introduced so that the SNR is improved. Fig. 9(e) shows the result of the STAP algorithm. Compared with the DBF algorithm, the spatial filter designed by the STAP algorithm does not produce a high gain of noise energy while filtering out the Doppler aliasing component, thus it has a lower noise level. Fig. 9(f) shows the result of the proposed algorithm. Compared with Fig. 9(a), the ghost targets are all effectively suppressed. Meanwhile, a higher SNR can be obtained since the result in Fig. 9(f) is obtained through spectral extrapolation and image fusion based on the result in Fig. 9(a) which already has a good SNR. Meanwhile, spectral extrapolation based on the minimum energy measure criterion also improves the SNR to some extent [21], as can be seen in Fig. 10. Fig. 10 compares the change of SNR in image domain (hereafter denoted as SNR_{out}) before and after spectral extrapolation using the simulation parameters from Table I, and the SNR_{in} is set from -100 to 45 dB. It can be seen from Fig. 10 that when the SNR_{in} is between -70 and 0 dB, the SNR_{out} after extrapolation is better than that before extrapolation, and the changes in the SNR_{out} before and after extrapolation are not obvious when the SNR_{in} is less than -70 dB or greater than 0 dB. This demonstrates that spectral extrapolation can improve the SNR to some extent.

To compare the various algorithms more quantitatively, imaging performance is evaluated and listed in Table II. It can

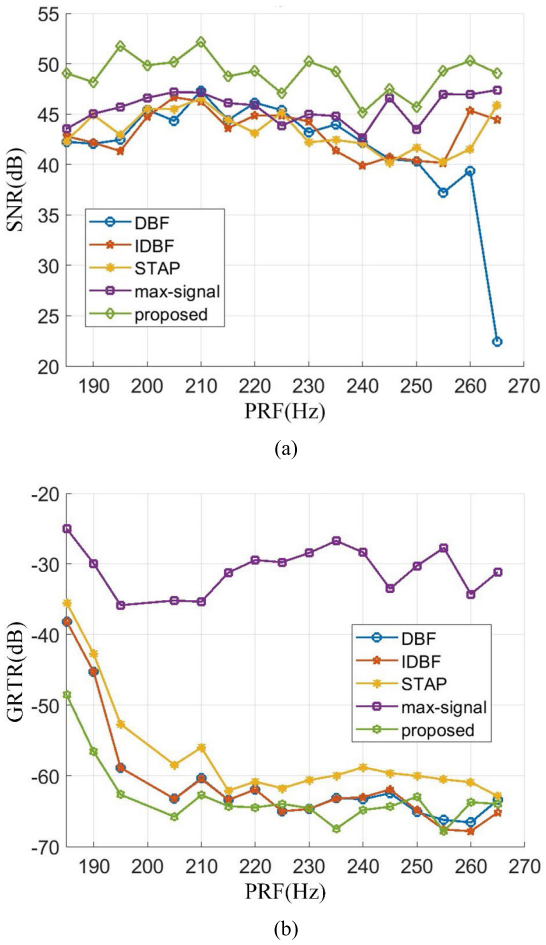


Fig. 11. Imaging performance of the different algorithms against varying PRF. (a) SNR. (b) GRTR.

be seen that the proposed algorithm has good comprehensive performance.

2) Imaging Performance Versus PRF: To further validate the robustness of the proposed algorithm, the four-channel SAR system is still considered whose system parameters are the same as those shown in Table I except for the PRF. The PRF takes values between 185 and 265 Hz at 5 Hz intervals, and other parameters are kept unchanged such that the different algorithms can be compared on the same basis. Fig. 11 shows the imaging performance of different algorithms versus PRF.

Fig. 11(a) demonstrates the variation of SNR_{out} with PRF for different algorithms. It can be seen that when the PRF is between 185 and 260 Hz, there is not much difference in the SNR of various types of algorithms, among which the best SNR performance is that of the proposed algorithm, followed by the max-signal algorithm. As the PRF gradually approaches the coinciding PRF, the SNR of the DBF algorithm decreases and deteriorates to below 25 dB at PRF of 265 Hz, while this phenomenon does not occur in the other algorithms because they all consider the suppression of the noise. Fig. 11(b) demonstrates the variation of GRTR with PRF for different algorithms. It can be seen that the max-signal algorithm has the worst GRTR performance because it effectively suppresses noise by maximizing the signal energy but fails to distinguish the signal energy of the ghost target and that of the real target.

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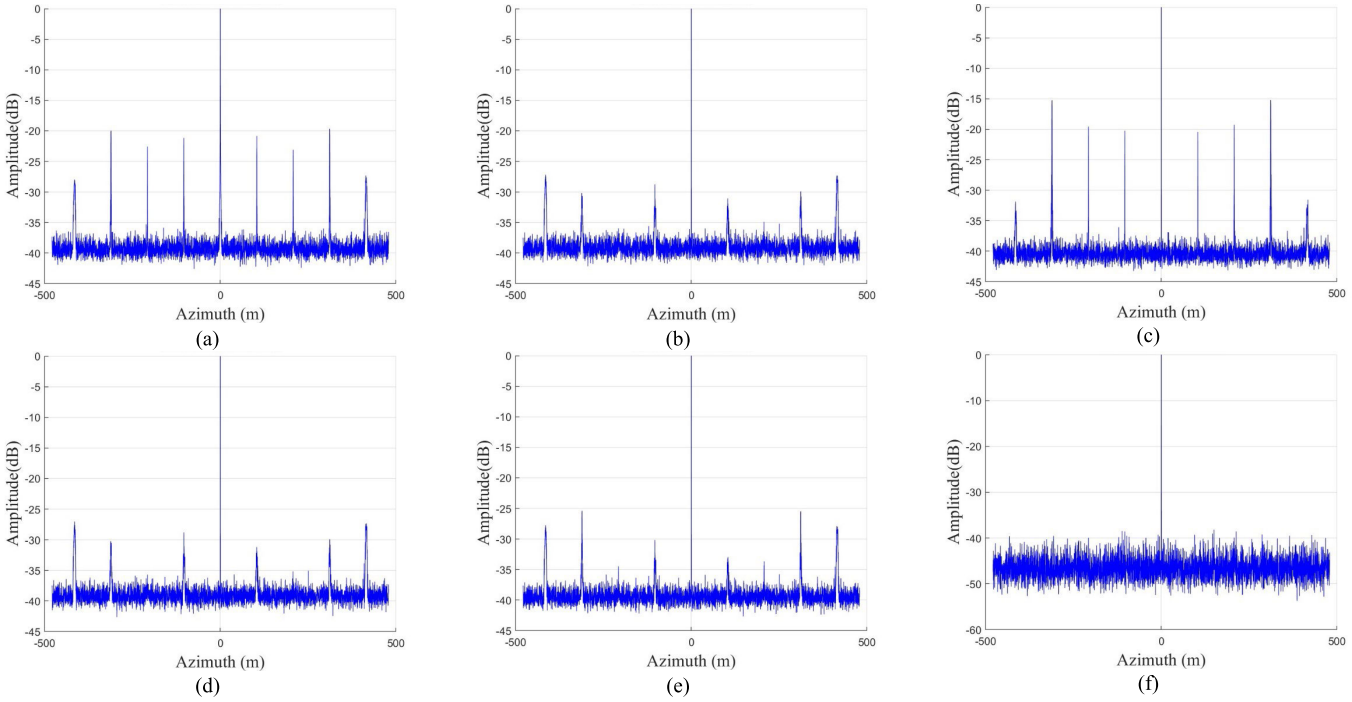


Fig. 12. Azimuth profile of different algorithms. (a) Periodic nonuniform sampling sequence processed by the chirp-scaling algorithm. (b) DBF algorithm. (c) Max-signal algorithm. (d) IDBF algorithm. (e) STAP algorithm. (f) Proposed algorithm.

In addition, the GRTR of the proposed algorithm is lower than those of the other algorithms when the PRF is lower than the uniform sampling frequency of 200 Hz. Under such PRF conditions, from the perspective of spatial domain filtering, the number of Doppler components containing signal energy is larger than the number of the channel. The reconstruction filters designed by other algorithms form at most $(M-1)$ zero-trapped positions in the spatial domain [11], which results in the inability to suppress the excess Doppler components that generate ghost targets. The proposed algorithm, however, utilizes the mechanism of ghost target generation under nonuniform sampling conditions and suppresses ghost targets by spectral extrapolation and image fusion, which is not limited by the number of Doppler aliasing components in the spatial domain, so it is still effective even when the PRF is lower than the uniform sampling frequency. Moreover, when the PRF is between 200 and 265 Hz, there is not much difference in the GRTR of several algorithms except the max-signal algorithm, and the proposed algorithm outperforms the DBF, the IDBF, and the STAP algorithms at most of the frequency points.

3) Imaging Performance at a Single PRF Under Nonbandlimited Condition: A single-point target simulation experiment based on the parameters in Table I is implemented. The PRF is set to be 230 Hz. The considered azimuth bandwidth is set to be 1380 Hz which is consistent with the analysis in Section III-A, and the sinc antenna pattern weighting is considered.

The results of different algorithms are shown in Fig. 12. To evaluate the superiority of the proposed algorithm, the imaging metrics are given in Table III. As Fig. 12 shows, in the results of the periodic nonuniform sampling sequence by chirp-scaling algorithm, four ghost targets occur on each side of the real target. In the result of max-signal algorithm, the ghost targets are almost not suppressed, while in the results of the DBF algorithm, the IDBF algorithm, and the STAP algorithm, the amplitude of ghost targets is suppressed to some extent, but there are still residual ghost targets. This is because these algorithms are developed for bandlimited signal, and the spectrum outside $[-M \cdot f_p/2, M \cdot f_p/2]$ introduces more spectral aliasing components which cannot be eliminated by these algorithms. The proposed algorithm, however, first obtains multiple subspectrums through spectral selection. Since the ratio of the spectral energy of each ghost target to that of the real target is different in different subspectra, the detection of all the ghost targets can be accomplished by utilizing this difference, and finally image fusion can be used to suppress all the ghost targets. Thus, the proposed algorithm can address the ghost target caused by the azimuth spectrum outside $[-M \cdot f_p/2, M \cdot f_p/2]$. In the result of the proposed algorithm, the ghost targets are significantly suppressed which demonstrates that the proposed algorithm has a better performance in suppressing the ghost targets than other algorithms under nonbandlimited condition.

TABLE III
EVALUATION OF IMAGING PERFORMANCE UNDER NONBANDLIMITED CONDITION

Algorithm	SNR (dB)	GRTR (dB)	Resolution (m)
DBF	33.5521	-38.1904	0.2323
IDBF	33.2405	-38.1344	0.2324
Max-signal	34.5291	-29.3504	0.2323
STAP	33.7607	-39.2048	0.2324
Proposed	34.7141	-44.3310	0.2243

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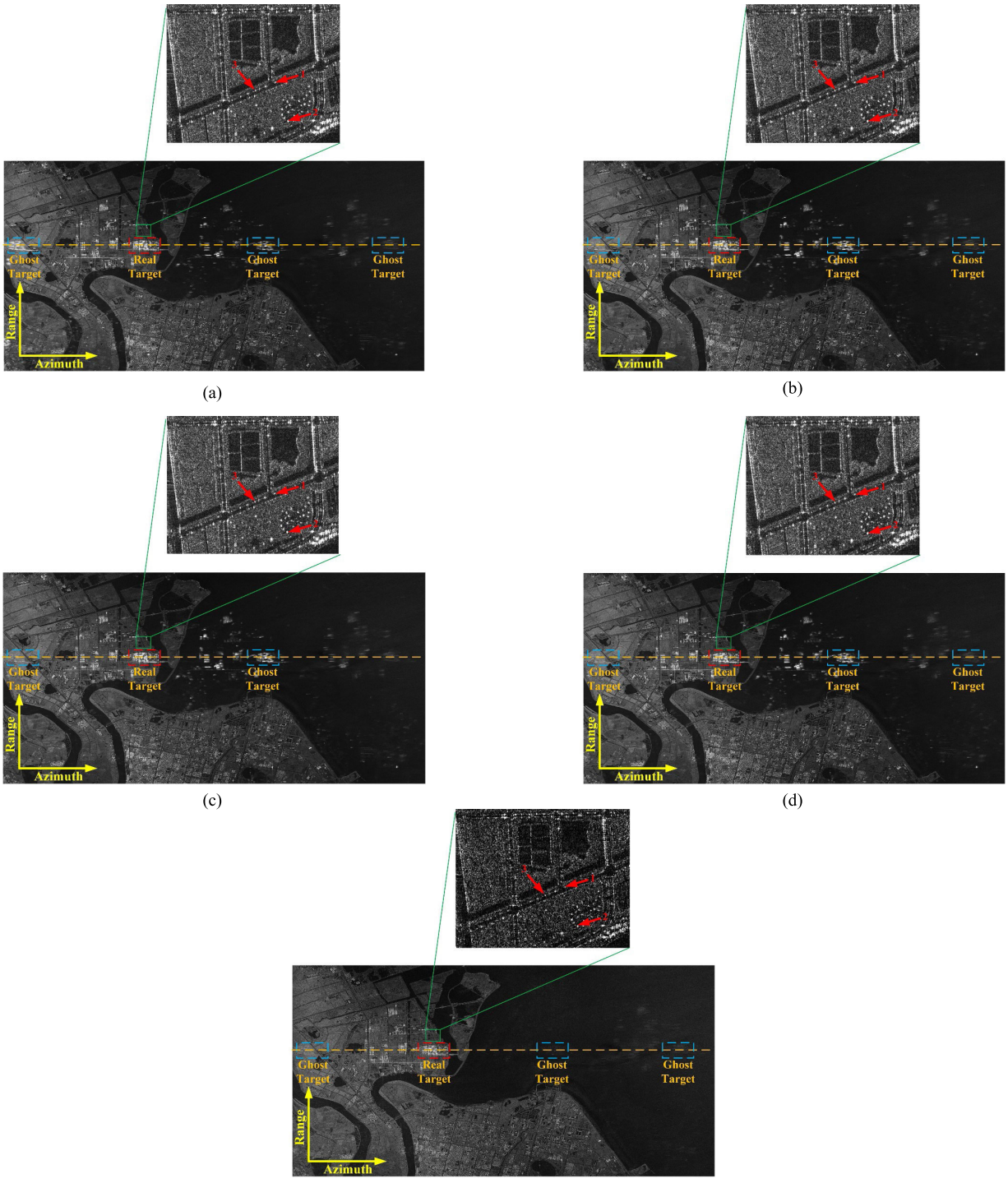


Fig. 13. Imaging results of different algorithms. (a) Max-signal algorithm. (b) DBF algorithm. (c) IDBF algorithm. (d) STAP algorithm. (e) Proposed algorithm.

B. Experiment for Real Multichannel Data

Real data obtained from the ultrafine Strip-map mode of the Gaofen-3 satellite are used to validate the performance of the proposed algorithm, with the imaging area being the land–sea interface with the Bohai Sea in Panjin City, Liaoning Province, China. The relevant parameters are listed in Table IV. The results of different algorithms are shown in Fig. 13, where the red dashed boxes indicate the real target area, and the blue dashed boxes indicate the ghost target distributed on each side of the real target.

It can be seen that in the result of the max-signal algorithm, there are obvious ghost targets. In the results of the DBF algorithm, the IDBF algorithm, and the STAP algorithm, the ghost targets on the left side of the real target are no longer visible, while the ghost targets on the right side is still obvious due to the weak energy of the sea surface. Compared with other algorithms, the proposed algorithm can effectively

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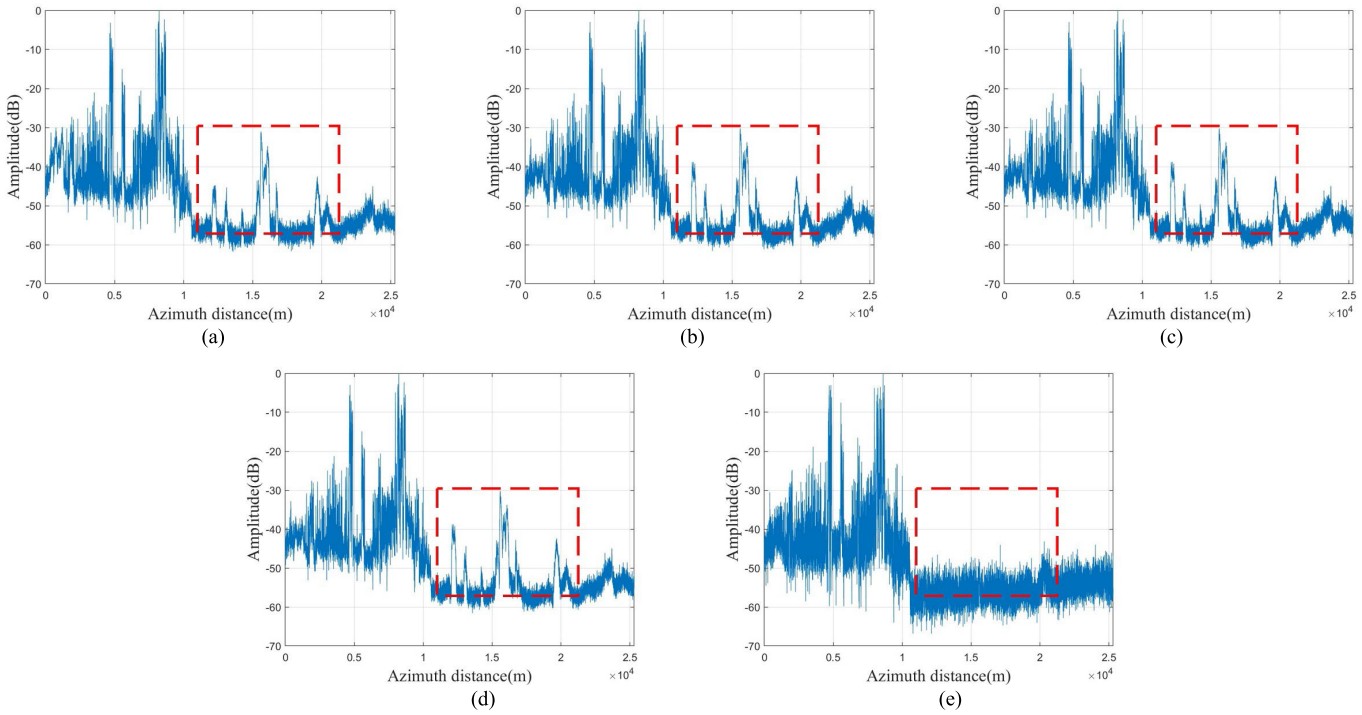


Fig. 14. Comparison of azimuth profiles at the yellow dashed line in Fig. 13. (a) Max-signal algorithm. (b) DBF algorithm. (c) IDBF algorithm. (d) STAP algorithm. (e) Proposed algorithm.

TABLE IV
SYSTEM PARAMETERS FOR REAL DATA

Parameter	Value
Carrier Frequency	5.4 GHz
Platform Velocity	6933 m/s
Pulse Repetition Frequency	2191 Hz
Sampling Rate	133.33 MHz
Bandwidth	100 MHz
Center Slant Range	846 km
Antenna Aperture Length	3.75 m
Channel Number	2

TABLE V
EVALUATION OF IMAGING PERFORMANCE OF REAL DATA EXPERIMENT

Algorithm	SNR (dB)	GRTR (dB)	Resolution (m)
DBF	39.7229	-30.9048	2.6268
IDBF	42.6358	-30.9036	2.6298
Max-signal	42.8131	-26.3538	2.5340
STAP	39.8113	-30.9036	2.6279
Proposed	43.0651	-37.3379	2.1292

suppress the ghost targets on each side of the real target, especially the ghost targets on the sea surface. As can be seen in Fig. 13(e), the ghost targets on the sea surface nearly disappear.

To better compare the degree of suppression for ghost targets, azimuth profiles of the ghost target region along yellow dashed lines in Fig. 13 are extracted and shown in Fig. 14. The red dashed box in Fig. 14 marks the azimuth profile corresponding to the ghost targets on the sea surface in Fig. 13. It can be seen that compared with the other algorithms, the proposed algorithm can suppress the ghost targets on the sea surface nearly completely.

Finally, the SNR, GRTR, and resolution of the results of different algorithms are given in Table V. There are several strong scattering point targets in the area marked by the green rectangular box in Fig. 13, and the three selected point targets for evaluating average azimuth resolution have been marked with red arrows. It can be seen that the proposed algorithm has a significant improvement in terms of SNR, GRTR, and resolution, compared to the other four algorithms.

V. CONCLUSION

In this article, the causes of the ghost target of azimuth multichannel SAR systems under nonuniform sampling conditions are first given in combination with the spectral characteristics of periodic nonuniform sampling sequence. Then a multichannel SAR imaging algorithm based on spectral extrapolation and image fusion is proposed. After simulation and real multichannel data verification, the proposed algorithm can suppress ghost targets effectively and obtain higher SNR compared with the conventional algorithm. In particular, the imaging resolution is also slightly improved due to the spectral extrapolation. Therefore, the proposed algorithm has a good comprehensive performance.

APPENDIX A

This section gives the relationship between the spectrum of $\mathbf{s}(\eta)$, which is denoted by $S(f_\eta)$, and the spectrum of $s_c(\eta)$ which is denoted by $S_c(f_\eta)$.

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Denoting the sampled value of signal $s_c(\eta)$ received by the mth channel at the lth pulse as $s_c(\eta_{m,l})$. Rearrange $s_c(\eta_{m,l})$ into the matrix shown in the following equation:

$$\begin{bmatrix} \text{echo}_1(\eta) \\ \text{echo}_2(\eta) \\ \vdots \\ \text{echo}_M(\eta) \end{bmatrix} = \begin{bmatrix} s_c(\eta_{1,1}) & s_c(\eta_{1,2}) & \cdots & s_c(\eta_{1,N}) \\ s_c(\eta_{2,1}) & s_c(\eta_{2,2}) & \cdots & s_c(\eta_{2,N}) \\ \vdots & \vdots & \ddots & \vdots \\ s_c(\eta_{M,1}) & s_c(\eta_{M,2}) & \cdots & s_c(\eta_{M,N}) \end{bmatrix} \tag{A1}$$

where $\text{echo}_m(\eta)$ denotes the sampling sequence corresponding to the received signal of the mth channel.

Firstly, $\text{echo}_m(\eta)$ is zero-padded and shifted as shown in (A2) to obtain the sequence $\text{echo}_m^{\text{zero}}(\eta)$

$$\begin{aligned} &\text{echo}_m^{\text{zero}}(\eta) \\ &= \Bigg[\underbrace{0, 0, \ldots, 0}_{(m-1)\text{zeros}}, s_c(\eta_{m,1}), \underbrace{0, 0, \ldots, 0}_{(M-1)\text{zeros}}, s_c(\eta_{m,2}), \\ &\quad \underbrace{0, 0, \ldots, 0}_{(M-1)\text{zeros}}, s_c(\eta_{m,3}), \ldots, \underbrace{0, 0, \ldots, 0}_{(M-1)\text{zeros}}, s_c(\eta_{m,N}) \Bigg]. \end{aligned} \tag{A2}$$

The relationship of $\mathbf{s}(\eta)$ and $\text{echo}_m^{\text{zero}}(\eta)$ is

$$\mathbf{s}(\eta) = \sum_{m=1}^{M} \text{echo}_m^{\text{zero}}(\eta). \tag{A3}$$

The nth sampling moment η_n of $\mathbf{s}(\eta)$ satisfies

$$\eta_n = n \cdot T - \Delta_n \tag{A4}$$

where Δ_n is the deviation between the nth sampling moment of $\mathbf{s}(\eta)$ and the ideal uniform sampling moment. T denotes the sampling interval under periodic uniform sampling condition.

According to (A3) and Fourier Transform, $S(f_\eta)$ can be derived by summing $\text{ECHO}_m^{\text{zero}}(f_\eta)$ which is the spectrum of $\text{echo}_m^{\text{zero}}(\eta)$

$$\begin{aligned} S(f_\eta) &= \sum_{m=1}^{M} \text{ECHO}_m^{\text{zero}}(f_\eta) \\ &= \sum_{m=1}^{M} \frac{1}{MT} \cdot \sum_{k=-\infty}^{+\infty} S_c(f_\eta - k \cdot f_p) \\ &\quad \cdot \exp\left[-j \cdot 2\pi \cdot (f_\eta - k \cdot f_p) \cdot t_m\right] \\ &\quad \cdot \exp\left[-j \cdot 2\pi \cdot (m-1) \cdot f_\eta \cdot T\right] \end{aligned} \tag{A5}$$

where $t_m = (m-1)d/2v$ which denotes the delay of the starting sampling moment of the mth channel relative to the starting sampling moment of the reference channel.

Substituting the r_m defined in (2) into (A5), $S(f_\eta)$ can be obtained

$$\begin{aligned} S(f_\eta) &= \frac{1}{T} \cdot \sum_{k=-\infty}^{+\infty} \left[\frac{1}{M} \cdot \sum_{m=1}^{M} F_{k,m}(f_\eta) \right] \cdot S_c(f_\eta - k \cdot f_p) \\ &= f_p \cdot \sum_{k=-\infty}^{+\infty} \left[\sum_{m=1}^{M} F_{k,m}(f_\eta) \right] \cdot S_c(f_\eta - k \cdot f_p) \end{aligned} \tag{A6}$$

where

$$\begin{aligned} F_{k,m}(f_\eta) &= \exp\left[-j \cdot 2\pi \cdot (f_\eta - k \cdot f_p) \cdot r_m \cdot T\right] \\ &\quad \cdot \exp\left[-j \cdot 2\pi \cdot k \cdot (m-1) \cdot \left(\frac{2\pi}{M}\right)\right]. \end{aligned} \tag{A7}$$

APPENDIX B

This section gives the positions of the ghost targets along azimuth direction away from the real target after matched filtering of the periodic nonuniform sampling sequence.

The SAR signal $s_c(\eta)$ can be equivalent to a chirp signal in the azimuth direction with linear frequency modulation properties. The Doppler center frequency is ignored for the convenience of the derivation, which does not affect the analysis that follows. Thus, $s_c(\eta)$ can be expressed as

$$s_c(\eta) = \text{rect}\left(\frac{\eta}{T_a}\right) \cdot \exp\left(-j \cdot \pi \cdot K_a \cdot \eta^2\right) \tag{B1}$$

where "rect(·)" denotes the rectangular function, T_a denotes synthetic aperture time, K_a denotes Doppler modulation frequency, and η denotes azimuth slow time.

The spectrum of $s_c(\eta)$ in (B1) can be expressed as

$$S_c(f_\eta) = \text{rect}\left(\frac{f_\eta}{B_a}\right) \cdot \exp\left(j \cdot \pi \cdot \frac{f_\eta^2}{K_a}\right). \tag{B2}$$

The periodic nonuniform sampling sequence $\mathbf{s}(\eta)$ is now processed using the matched filter $H(f_\eta)$ shown in the following equation:

$$H(f_\eta) = \text{rect}\left(\frac{f_\eta}{f_s}\right) \cdot \exp\left(-j \cdot \pi \cdot \frac{f_\eta^2}{K_a}\right). \tag{B3}$$

According to (A6) and (B3) the frequency-domain result of matched filtering is given as

$$\begin{aligned} \hat{S}(f_\eta) &= S(f_\eta) \cdot H(f_\eta) \\ &= f_p \cdot \sum_{k=-\infty}^{+\infty} \left[\sum_{m=1}^{M} F_{k,m}(f_\eta) \right] \cdot \text{rect}\left(\frac{f_\eta - k \cdot f_p}{B_a}\right) \\ &\quad \cdot \exp\left[+j \cdot \pi \cdot \frac{(f_\eta - k \cdot f_p)^2}{K_a}\right] \cdot \text{rect}\left(\frac{f_\eta}{f_s}\right) \\ &\quad \cdot \exp\left(-j \cdot \pi \cdot \frac{f_\eta^2}{K_a}\right) \\ &= M \cdot \text{rect}\left(\frac{f_\eta}{B_a}\right) + f_p \cdot \sum_{m=1}^{M} F_{k,m}(f_\eta) \\ &\quad \cdot \sum_{l=0}^{1} \sum_{k=1}^{M-1} \left[B_{k,l} \cdot \exp\left(\frac{j \cdot \pi \cdot \phi_{k,l}^2}{K_a}\right) \right. \\ &\quad \left. \cdot \exp\left(-j \cdot 2\pi \cdot \frac{\phi_{k,l}}{K_a}\right) \right] \end{aligned} \tag{B4}$$

where

$$\phi_{k,l} = k \cdot f_s / M - l \cdot f_s \tag{B5}$$

$$B_{k,l} = \text{rect}\left[\frac{f_\eta - \frac{1}{2} \cdot \phi_{k,l} - \frac{1}{4} \cdot (-1)^l \cdot (f_s - B_a)}{(\frac{B_a + f_s}{2}) + (-1)^l \cdot \phi_{k,l}} \right]. \tag{B6}$$

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The time-domain results of the matched filtering can be obtained by performing inverse Fourier Transform of (B4)

$$\begin{aligned}\hat{s}(\eta) &= M \cdot B_a \cdot \sin c(B_a \eta) \\ &+ f_p \cdot \sum_{m=1}^{M} f_{k,m}(\eta) \otimes \sum_{l=0}^{1} \sum_{k=1}^{M-1} \exp\left(\frac{j \cdot \pi \cdot \phi_{k,l}^2}{K_a}\right) \\ &\cdot \left[\left(\frac{B_a + f_s}{2}\right) + (-1)^l \cdot \phi_{k,l}\right] \\ &\cdot \sin c\left\{\left[\left(\frac{B_a + f_s}{2}\right) + (-1)^l \cdot \phi_{k,l}\right] \cdot \left(\eta - \frac{\phi_{k,l}}{K_a}\right)\right\} \\ &\cdot \exp\left\{j \cdot 2\pi \cdot \left[\frac{1}{2} \cdot \phi_{k,l} + \frac{1}{4} \cdot (-1)^l \cdot (f_s - B_a)\right]\right. \\ &\left. \cdot \left(\eta - \frac{\phi_{k,l}}{K_a}\right)\right\} \end{aligned} \tag{B7}$$

where "$\otimes$" denotes convolution operation, and $f_{k,m}(f_\eta)$ is the inverse Fourier Transform of $F_{k,m}(f_\eta)$. The first term corresponds to the real target and the second term corresponds to each ghost target, and it can be seen that the ghost target corresponds to the azimuthal moment

$$T_{k,l} = (l \cdot \alpha_s - k \cdot \alpha_s / M) \cdot T_a \tag{B8}$$

where $l = 0, 1, k = 1, 2, \ldots, M-1$, T_a denotes the synthetic aperture time, and $\alpha_s = M \cdot f_p / B_a$. Therefore, the distance that each ghost target deviates from the real target along azimuth direction is

$$\mathrm{POS}_{k,l} = v \cdot T_{k,l} = (l \cdot \alpha_s - k \cdot \alpha_s / M) \cdot v \cdot T_a. \tag{B9}$$

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